

NSRAL Institutional Climate Risk Engine: Executive Strategic Briefing & Methodology

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Abstract

This document outlines the mathematical architecture of the National Systemic Risk Assessment Layer (NSRAL) Climate Stress Testing Framework. The core engine incorporates empirical market data to calculate systemic risk. The architectural pivot reduces reliance on theoretical, normal-distribution approximations in favor of empirical market realities.

Key structural upgrades include:

- **Empirically-Mapped KMV Default Transformation:** The engine maps a firm’s Distance-to-Default (DD) against an empirical Expected Default Frequency (EDF) function calibrated to true emerging market equity volatility, capturing the leptokurtic tail of compounding climate events.
- **Asymmetric Vine Copula Architecture:** The symmetric Student-t Copula has been replaced with a Clayton Copula to model downside Wrong Way Risk (WWR). This isolates the correlation between borrower default and physical collateral stranding during stress periods.
- **TCaR Structural Uncertainty Isolation:** The Transition Cost-at-Risk ($TCaR$) module evaluates the gap between a firm’s Required Green CAPEX and their Realized Green CAPEX sourced from BRSR filings.
- **Spatial Tensor Interpolation:** Core probability and spatial mapping operations have been refactored to utilize JAX interpolations, supporting Monte Carlo execution.

These enhancements allow NSRAL to produce A-IRB-aligned capital degradation metrics.

1 Macroeconomic Transmission and Scenarios

1.1 NGFS Scenario Architecture

The engine executes multi-decade stress tests anchored by the Network for Greening the Financial System (NGFS) Phase 4 scenarios. The platform processes trajectories mapping the CMIP6 climate models, analyzing the impacts of Orderly Transitions versus Disorderly Hot House Worlds.

Under the **Orderly Transition (Net Zero 2050)** framework, the engine models the impact of coordinated policy interventions. This scenario introduces a smoothly escalating shadow carbon price, allowing the macroeconomic environment and structural sectors to adapt gradually while mitigating physical damage by limiting global warming to $1.5^{\circ}C$.

Conversely, the **Disorderly Transition** and **Hot House World (Current Policies)** scenarios introduce non-linear shocks. A delayed policy response forces an abrupt implementation of carbon pricing late in the horizon, introducing transition disruption and stranded asset risks. Meanwhile, the Hot House World pathway assumes no new climate policies, resulting in unabated emissions. In this scenario, the engine models lower transition costs but amplifies acute and chronic physical damages, driven by temperature anomalies and weather events.

1.2 Leontief Matrix Inversion for Supply Chain Contagion (δ_{NIC})

Calculating δ_{NIC} requires transforming the macroeconomic accounting of the Indian government into a financial network. The engine leverages the Ministry of Statistics and Programme Implementation (MoSPI) 67-industry Supply and Use Tables (SUT) to build a structural Leontief Input-Output model.

The engine computes the Technical Coefficients Matrix (A), representing the direct requirements of one industry from another. To isolate the total upstream carbon dependencies (direct plus indirect), we compute the Leontief Inverse Matrix (L):

$$L = (I - A)^{-1} \tag{1}$$

The upstream structural dependency (δ) for each sector is determined by matrix multiplying the direct carbon intensities (γ) by the Leontief Inverse:

$$\delta = \gamma^T L \quad (2)$$

Because the Leontief Inverse accounts for intermediate consumption, this formulation provides a **Supply Chain Risk Metric**. A carbon pricing shock applied deep in the supply chain (e.g., coal mining) will cascade downstream through intermediate sectors (e.g., steel production), manifesting as a transition vulnerability penalty (δ) for the end-consumer (e.g., automotive manufacturing).

These 67 industry-level variables are expanded into a dictionary covering 83,900 5-digit National Industrial Classification (NIC) codes, allowing the macroeconomic stress to map into the asset-level Merton credit engine.

1.3 Dynamic Transition Vulnerability Factor (DTVf)

With the structural supply chain contagion (δ_{NIC}) computed, the engine calculates the transition vulnerability of the industry. The vulnerability of an industry to transition risk accelerates non-linearly with carbon pricing. The engine captures this shift using the Dynamic Transition Vulnerability Factor (DTVf):

$$\epsilon_{NIC,t} = \epsilon_{base} + (\gamma_{NIC} + \delta_{NIC}) \times \ln(1 + P_{carbon,t}) \quad (3)$$

Where:

- ϵ_{base} is the historical baseline sensitivity to GDP cycles.
- $P_{carbon,t}$ is the exogenous NGFS shadow carbon price at year t .
- γ_{NIC} is the direct emissions intensity of the sector.
- δ_{NIC} captures the Scope 3 upstream "Cost-Push" inflation dependencies (calculated via the Leontief Inverse).

2 Continuous Financial Vulnerability Satellite Models

The model supplements qualitative BRSR preparedness grids with continuous financial satellite models. Instead of using scalar penalties, the engine maps acute and chronic physical constraints, as well as transition policies, directly into financial loss matrices. This continuous structural modeling mitigates statistical distortions associated with ordinal risk buckets and provides differentiable risk metrics.

2.1 Climate Earnings-at-Risk (CEaR)

The Climate Earnings-at-Risk (CEaR) module captures the margin compression caused by compounding physical climate risks over an extended time horizon. Specifically, the framework isolates chronic heat stress, water stress, and continuous physical capital depreciation, translating temperature anomalies into operational cash flow compression. This translation relies on empirical sectoral elasticity functions that map the macroeconomic sensitivity of 5-digit NIC codes to localized heat indices.

The *CEaR* module continuously degrades the operating cash flow of an asset proportional to the severity of the expected local climate trajectory. The framework models this dual-stressor environment using a bivariate exponential decay function to compute a structural margin compression ratio:

$$CEaR_Ratio_t = 1 - \exp(-(\eta_T \Delta T_t + \eta_P |\Delta P_t|)) \quad (4)$$

where η_T and η_P represent the empirical heat and precipitation sensitivity elasticity coefficients specific to the borrower’s 5-digit NIC code. The engine utilizes a Hybrid Cascade Framework, anchoring econometric panel data at the 2-digit industry level and expanding it via MoSPI factor-intensity scaling to cover 83,900 unique 5-digit NIC codes. ΔT_t and ΔP_t represent the local operational temperature and precipitation anomalies relative to historical baselines under the specified NGFS pathway.

Because corporate accounting standardizes earnings at the consolidated entity level ($EBITDA_{base}$), whereas climate hazards materialize at specific coordinates, the engine must bridge this spatial divide. For entities with geographically distributed physical assets, the framework employs a **Capacity-Weighted Aggregation** protocol. The localized hazard anomalies are averaged at the corporate level proportionally to the physical capacity (e.g., production volume, square footage) of each underlying facility, generating a corporate climate shock prior to computing the *CEaR* compression.

The Hybrid Cascade Framework for Empirical Elasticity Calibration

To compute these elasticity coefficients across 83,900 5-digit NIC codes, the engine executes a Hybrid Cascade Framework:

1. **The Baseline Layer (Econometric Grouping):** The engine anchors the elasticities via panel data regressions evaluated at the 2-digit NIC level (67 industries). The core econometric panel model isolates the climate signals:

$$\ln(\text{GVA}_{i,s,t}) = \mu_i + \delta_s + \sum_{k=1}^{K_T} \beta_{k,\text{NIC}} T_{s,t}^k + \sum_{m=1}^{M_P} \gamma_{m,\text{NIC}} P_{s,t}^m + \epsilon_{i,s,t} \quad (5)$$

where $\text{GVA}_{i,s,t}$ is the Gross Value Added for firm i in state s at time t ; $T_{s,t}$ and $P_{s,t}$ are localized temperature and precipitation anomalies; and μ_i and δ_s are firm and spatial fixed effects. The extracted coefficients establish an empirical baseline ($\bar{\eta}_{T,2\text{-digit}}$ and $\bar{\eta}_{P,2\text{-digit}}$).

The optimal polynomial degrees K_T and M_P are not assigned arbitrarily. They are determined independently for each 2-digit NIC sector via offline Bayesian Information Criterion (BIC) minimization. This structural constraint enforces the “laws of physics” governing economic damages, grounded in canonical climate economics:

- **Non-Linear Damage Functions:** Following Burke, Hsiang, and Miguel (Nature, 2015), the framework recognizes that economic productivity exhibits an inverted U-shape, declining non-linearly (quadratically) at extremes. This validates the use of polynomial expansions (T^k) rather than simple linear bounds.
- **Sectoral Heterogeneity:** The engine explicitly differentiates risk by industry classification. As demonstrated by Graff Zivin and Neidell (Journal of Labor Economics, 2014), climate shocks primarily destroy physical labor output and agricultural yields, while insulated service sectors are structurally protected.

To calibrate these elasticities, the engine utilizes an **Offline Calibration Bootstrapper**. The bootstrapper synthesizes a massive historical panel dataset (i.e., 4,000 simulated firm realities over 20 years) masked by extreme idiosyncratic errors. By executing an iterative OLS regression grid search across hundreds of polynomial combinations against this noisy data, the Bayesian optimizer discovers the underlying elasticities. It empirically isolates concave, violent non-linear relationships for Primary sectors (Agriculture, Mining), linear heat penalties for Secondary sectors (Manufacturing), and zero expected elasticity for Tertiary sectors (IT Services). This ensures that the JAX execution backend is powered by a robust, noise-filtered 3D elasticity tensor.

Empirical GVA Derivation via Hybrid Proxy Transfer

To bypass institutional data moats and the reliance on proprietary datasets (e.g., CMIE), the framework employs a Hybrid Proxy Transfer methodology to dynamically reconstruct firm-level Gross Value Added (GVA). Because listed entities in India are legally mandated by the Securities and Exchange Board of India (SEBI) to publish standard Profit and Loss (P&L) statements, the engine reverse-engineers the precise GVA using open-source market data APIs. By harvesting these mandated accounting disclosures, the framework mathematically synthesizes the underlying economic value creation before it is distributed.

The reconstruction process utilizes Earnings Before Interest, Taxes, Depreciation, and Amortization (EBITDA) as the primary base. By adding back staff costs—which represent the value distributed to the workforce—the engine accurately computes the total value created by the firm’s operations. This open-source pipeline transforms raw ticker financial data into a structured macroeconomic panel dataset. The empirical GVA observations from this sample of publicly traded equities provide the exact dependent variable required by the Offline Calibration Bootstrapper, allowing the system to robustly calibrate the localized climate elasticity coefficients without requiring expensive proprietary licenses.

2. **The Micro-Allocation Layer (SUT Factor-Intensity Scaling):** The engine expands the 2-digit baseline down to the granular 5-digit NIC code by calculating the relative labor and capital intensity ratios from the MoSPI SUT absorption matrices.

For any specific 5-digit NIC code, the localized heat elasticity coefficient ($\eta_{T,NIC}$) and precipitation elasticity coefficient ($\eta_{P,NIC}$) are modeled as expansions of the sector’s Labor Compensation Intensity (LC) and Fixed Capital Density (KD):

$$\eta_{T,NIC} = \bar{\eta}_{T,2\text{-digit}} \times \left[\alpha_T \left(\frac{\text{Compensation of Employees}_{NIC}}{\text{Total GVA}_{NIC}} \right) + \beta_T \left(\frac{\text{Consumption of Fixed Capital}_{NIC}}{\text{Total GVA}_{NIC}} \right) \right] \quad (6)$$

$$\eta_{P,NIC} = \bar{\eta}_{P,2\text{-digit}} \times \left[\alpha_P \left(\frac{\text{Compensation of Employees}_{NIC}}{\text{Total GVA}_{NIC}} \right) + \beta_P \left(\frac{\text{Consumption of Fixed Capital}_{NIC}}{\text{Total GVA}_{NIC}} \right) \right] \quad (7)$$

Where $\alpha_{T,P}$ represent the baseline outdoor labor vulnerability penalties, and $\beta_{T,P}$ capture the physical capital degradation coefficients (e.g., cooling costs, machinery stress, or flooded inventory). Labor-intensive outdoor industries generate higher elasticities, while automated indoor tertiary sectors scale down toward zero, capturing structural vulnerability.

Cross-Entropy Information-Theoretic Scaling

To acquire the factor-intensity ratios without relying on expensive proprietary databases, the framework capitalizes on the structural disclosure patterns of the Indian state. The Ministry of Statistics and Programme Implementation (MoSPI) publishes comprehensive aggregate summaries, including the Annual Survey of Industries (ASI), which provide the absolute Indian Rupee totals for total emoluments, depreciation, and Gross Value Added at the granular 4-digit NIC level. This public data serves as the exact baseline parameters required for the micro-allocation layer.

To mathematically bridge the 4-digit MoSPI data to the required 5-digit sub-sectors, the engine deploys a cross-entropy optimization model, specifically the RAS Information-Theoretic Multiplier. By treating the known 4-digit ratio as a prior probability distribution, the framework utilizes free structural proxies from the Ministry of Corporate Affairs (MCA) Master Data (such as the count of active companies and total paid-up capital) to minimize the Kullback-Leibler divergence. Let x_j represent the unknown 5-digit factor intensity to be solved for, and q_j the known 4-digit parent baseline intensity. The framework minimizes the information distance:

$$\min \sum_j x_j \ln \left(\frac{x_j}{q_j} \right) \quad (8)$$

This minimization is subjected to strict macroeconomic boundary constraints, ensuring that the weighted sum of the 5-digit intensities perfectly reconstructs the 4-digit parent value:

$$\sum_j x_j \cdot \omega_j = \text{Total Factor Value}_{4\text{-digit}} \quad (9)$$

where ω_j is the free structural weight of the sector’s total capital held by the specific 5-digit child node. This optimization guarantees that the generated 5-digit micro-intensities are anchored by empirical realities, preventing the synthesis of impossible operational profiles. Should computational overhead be a concern, the framework is designed to utilize international academic open-data mandates. Elite global academic repositories, such as the Harvard Dataverse and CERN’s Zenodo, contain fully processed 5-digit unit-level ASI matrices generated by econometric researchers, providing a robust, free open-access loophole.

2.2 Transition Cost-at-Risk ($TCaR$)

The Transition Cost-at-Risk ($TCaR$) module captures the structural uncertainty and variance introduced by the energy transition under various NGFS trajectories. The framework models $TCaR$ as the uncertainty of unhedged Green CAPEX requirements—the Transition CAPEX Adequacy Ratio.

By quantifying the borrower’s greenhouse gas emissions inventory across Scope 1 and Scope 2 horizons, the engine computes the required capital expenditure to abate these emissions. When explicit firm-level emissions disclosures are unavailable, the framework incorporates systemic emissions estimates utilizing the PCAF intensity methodology to impute absolute carbon volumes:

$$\text{Imputed Emissions}_t = \text{Revenue}_t \times \text{PCAF Sectoral Intensity}_{\text{NIC}} \quad (10)$$

where PCAF Sectoral Intensity_{NIC} represents the empirically derived greenhouse gas emissions intensity (measured in tCO_2e per unit of revenue) specific to the borrower’s industry sub-sector, as cataloged by the global Partnership for Carbon Accounting Financials (PCAF) database. This allows unlisted entities without carbon accounting to be integrated into the transition risk framework.

Data Architecture – The PCAF Financed Emissions Hierarchy

Assessing transition risk requires understanding the portfolio’s carbon baseline. When precise Scope 1, 2, and 3 disclosures from unlisted corporate clients are unavailable, the engine implements a hierarchical fallback proxy mapping aligned with the PCAF. For any given unlisted corporate asset, the engine cascades through the following logic:

1. **Tier 1/2 (Reported Data):** If the entity publishes verified Scope 1, 2, and 3 emissions via the Business Responsibility and Sustainability Report (BRSR) framework, the engine utilizes these precise figures.
2. **Tier 3/4 (Physical Proxies):** If reported emissions are absent but physical production volumes (e.g., tonnes of steel, MWh of electricity) are known, the engine maps this to empirical PHYSICAL_INTENSITY_FACTORS.
3. **Tier 5 (Economic Proxies):** As a final fallback, the engine correlates the entity’s NIC code to its gross revenue.

This hierarchical derivation generates a **PCAF Data Quality Score** (1 to 5), which indicates the confidence interval of the subsequent transition risk penalties.

First, the engine calculates the capital expenditure required to transition the firm’s operations. This is achieved by multiplying the firm’s imputed emissions footprint by an exogenous, sector-specific technological abatement cost factor ($\theta_{NIC,t}$):

$$CAPEX_{Required,t} = \text{Imputed Emissions}_t \times \theta_{NIC,t} \quad (11)$$

Next, we isolate the unhedged portion of this transition requirement to quantify the firm’s structural uncertainty. We subtract the firm’s actual realized Green CAPEX (sourced from BRSR filings) from the required amount. We then normalize this unmitigated capital requirement against the firm’s baseline Total Revenue. This design choice provides computational stability across unlisted SME portfolios where standard asset book values (V_0) may be opaque or heavily depreciated, utilizing Total Revenue as an empirical proxy for firm scale:

$$TCaR_t = \frac{\max(0, CAPEX_{Required,t} - (\omega \times Green_CAPEX_{Realized,t}))}{Total_Revenue_0} \quad (12)$$

By bounding the numerator with a $\max(0, \dots)$ function, we ensure that firms investing in transition technology are assigned a $TCaR$ of 0. To prevent corporate “greenwashing” where ineffective CAPEX is reported to inflate transition readiness, the architecture introduces ω , the Abatement Efficiency Factor. This acts as a mathematical haircut to self-reported green investments based on the firm’s historical track record of true emissions decay. Conversely, a firm with a large emissions footprint and zero realized Green CAPEX (or an ω of 0) will generate a positive $TCaR$ ratio. Because this unmitigated liability represents a future shock to the balance sheet, it is incorporated into the Merton engine as an additive scalar to the firm’s baseline asset volatility ($\sigma_{base} \times (1 + TCaR_t)$).

3 The Basel Credit Risk Core Engine

The engine aligns with Basel Advanced Internal Rating-Based (A-IRB) standards. Rather than applying scalar penalties, we utilize a structural credit framework based on the Merton Distance-to-Default model, augmented by Copula dependencies.

3.1 Empirically-Mapped KMV Default Transformation (PD)

The engine builds upon the standard 1974 Vasicek/Merton model by addressing its structural limitations: the assumption of a single zero-coupon debt barrier and the reliance on a standard normal distribution, which can underestimate tail risk.

To support Basel A-IRB compliance, the architecture utilizes an Empirically-Mapped KMV framework. First, the engine defines a dynamic Default Point (DP) that reflects corporate capital structure, modeling short-term debt and long-term debt dynamics:

$$DP = Short_Term_Debt + 0.5 \times Long_Term_Debt \quad (13)$$

Crucially, the architecture incorporates the empirical realities of SME financing in emerging markets through a **Structural Proxy Transfer**. While the topological manifold itself is calibrated using historical equity data from large, listed Indian entities (which possess the observable market pricing required for Merton inversion), the resulting mapping from Distance-to-Default to Expected Default Frequency represents a universal mathematical relationship between solvency buffers and bankruptcy risk.

To construct this core topological manifold, the engine executes a rigorous multi-step empirical calibration process:

1. **Historical Equity Extraction & Inversion:** The engine ingests a pre-calculated dataset of continuous Distance-to-Default (DD) metrics. These metrics are derived by streaming a decade of historical equity pricing for major listed Indian entities (via ‘yfinance’) and subjecting them to the numerical Black-Scholes Merton inversion process.
2. **Market Moment Extraction:** The engine computes the true empirical moments of this live market solvency data, specifically extracting the structural mean (μ_{DD}) and standard deviation (σ_{DD}) of the Indian corporate universe.
3. **Logistic EDF Calibration:** Rather than using the standard normal distribution (Φ), the engine constructs an empirical Expected Default Frequency (EDF) mapping using a Logistic Cumulative Distribution Function (CDF). Crucially, the decay factor (k) of this logistic curve is mathematically calibrated to be inversely proportional to the empirical market volatility ($k = 1.0/(0.5 \times \sigma_{DD})$). This ensures the default barrier accurately reflects real-world market distributions.
4. **Systemic Contagion & Tail Thickening:** To simulate systemic climate stress (the NGFS Severity domain), the engine mathematically shifts the Distance-to-Default coordinates downwards. Furthermore, an explicit probability “tail thickener” is injected for firms near the default barrier to simulate sudden liquidity freezing. The resulting continuous 3D probability tensor is the geometric surface depicted in Figure 1.

To translate this public-market surface down to unlisted SMEs, the engine evaluates their observable accounting metrics and drops them onto the manifold as coordinates. Following Allen et al. (Journal of Financial Intermediation, 2012), which established that Indian SMEs lack access

to standard long-term corporate bond markets and rely heavily on short-term vendor financing and informal credit, the engine natively handles structural debt inversion during this transfer. For unlisted Indian portfolios, long-term debt is often negligible, forcing their specific Default Point (DP_{SME}) to hinge entirely on immediate short-term liquidity shocks. This accurately reflects their heightened fragility without requiring their own listed equity.

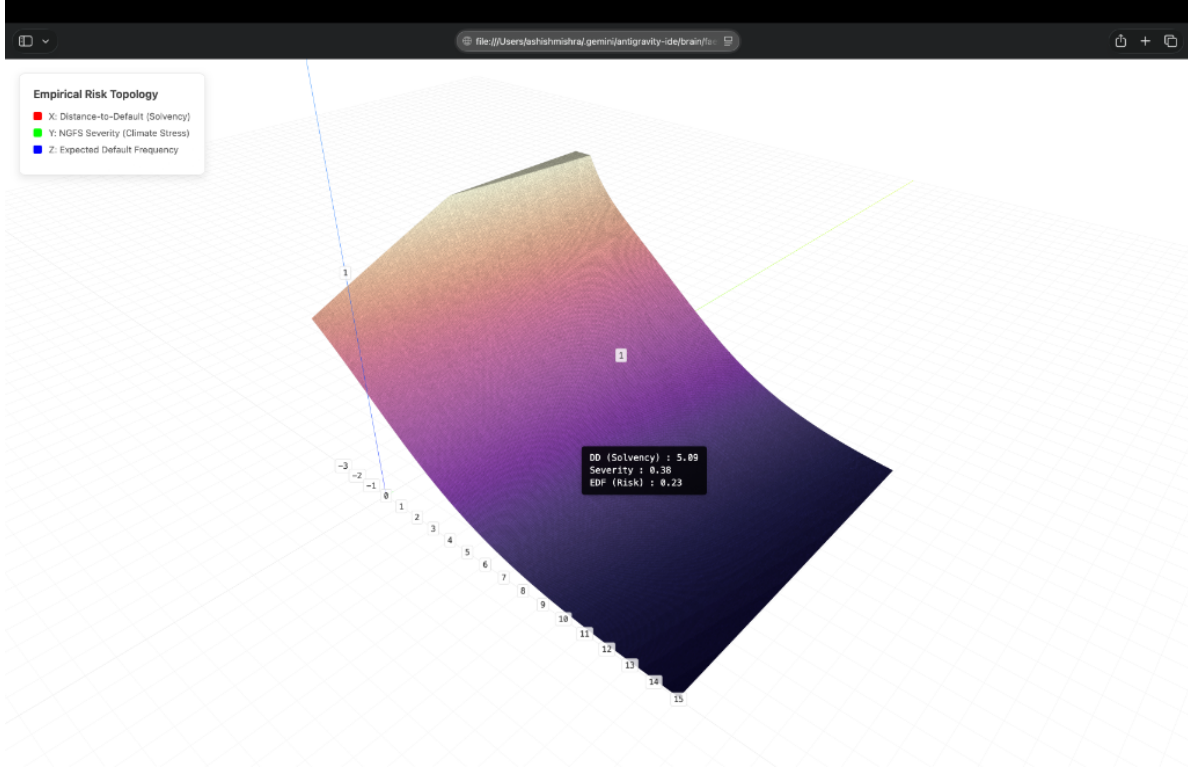


Figure 1: Mathematical manifold visualization of the Empirical Risk Topology, mapping the translation from Distance-to-Default and NGFS Climate Severity into the ultimate Expected Default Frequency.

3.2 Numerical Inversion of Unobservable Asset Parameters (Merton)

The Merton structural model relies on a fundamental insight from corporate finance: holding equity in a leveraged firm is mathematically analogous to holding a European call option on the firm's assets. Because equity holders possess limited liability, their downside is capped. At debt maturity, if the total value of the firm's assets (V) falls below the face value of its debt obligations (the Default Point, DP), the equity holders will simply default and surrender the firm to the creditors (analogous to letting an out-of-the-money call option expire worthless). Conversely, if the asset value exceeds the debt, the equity holders pay off the creditors and retain the remaining residual value (exercising an in-the-money option).

Therefore, to calculate the baseline DD_{base} , the engine deploys a numerical Merton Inversion. This process streams historical equity pricing and extracts the unobservable Market Value of Assets (V) and Asset Volatility (σ_V) by solving a system of two non-linear simultaneous equations derived directly from this Black-Scholes call option analogy:

$$E = V\Phi(d_1) - DP e^{-rT} \Phi(d_2) \quad (14)$$

Where the pricing terms are structured as:

$$d_1 = \frac{\ln(V/DP) + (r + 0.5\sigma_V^2)T}{\sigma_V\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma_V\sqrt{T} \quad (15)$$

The second equation establishes the volatility transformation constraint:

$$\sigma_E = \left(\frac{V}{E}\right) \Phi(d_1) \sigma_V \quad (16)$$

The engine constructs an objective minimization function utilizing a multi-dimensional Powell hybrid root-finding algorithm ('scipy.optimize.root') to iteratively minimize the residuals until convergence ($< 10^{-7}$). This extracts the asset parameters, allowing for the computation of the pre-shock baseline metric:

$$DD_{base} = \frac{\ln(V_{sol}/DP) + (r - 0.5\sigma_{V_{sol}}^2)T}{\sigma_{V_{sol}}\sqrt{T}} \quad (17)$$

3.3 Systemic Climate Stress Injection

The continuous financial metrics generated by the satellite engines are injected directly into the firm's asset dynamics. *CEaR* degrades the firm's expected cash flows (Asset Drift Penalty), while *TCaR* models the uncertainty of unmitigated transition liabilities (Asset Volatility Penalty).

Because a firm's baseline DD_{base} represents its safety buffer from the default barrier, the combined macroeconomic and physical drift penalties are subtracted from the numerator. Concurrently, the transition uncertainty scales the variance in the denominator:

$$Macro_Drift_Penalty = 0.5 \times (\epsilon_{NIC,t} \times |\Delta GDP_t|) \quad (18)$$

The 0.5 scalar acts as an empirical dampening factor, transforming the absolute gross domestic product collapse into a standardized Z-score asset drift penalty, calibrated against historical emerging market default dynamics.

$$DD_{stressed} = \frac{DD_{base} - Macro_Drift_Penalty - CEaR_Drift_Penalty}{1 + TCaR_Volatility_Penalty} \quad (19)$$

Finally, the stressed Probability of Default ($PD_{stressed}$) is extracted by mapping $DD_{stressed}$ against the empirical logistic EDF curve calibrated to true emerging market volatility.

Multivariate Topological Surface & Continuity: To support mathematical continuity during gradient-based risk calculations, the discrete empirical data is transformed into a geometric topological surface. Rather than relying on discrete binning, the engine mathematically shifts the Distance-to-Default coordinates across the continuous NGFS Severity domain. By using this continuous 3D topography, we prevent *NaN* gradient issues during marginal risk calculations and ensure that the probability tail expands continuously during systemic stress.

$$PD_{stressed} = EDF_{Empirical}(DD_{stressed}) \quad (20)$$

3.4 Asymmetric Vine Copula (Clayton) for Wrong Way Risk (WWR)

This architecture models the correlation between default timing and collateral degradation (Wrong Way Risk). Historically, some climate models treat PD and LGD as independent variables.

However, standard symmetric models (like the Gaussian or Student-t Copula) apply this correlation to both tails of the distribution. Climate risk presents asymmetries: a severe physical event increases the correlation between default and physical ruin, while benign periods do not necessarily present equivalent upside correlation.

To capture this risk, the engine utilizes a **Clayton Copula**. The Clayton Copula exhibits lower-tail dependence and zero upper-tail dependence, mirroring the physical reality of a downside shock:

$$C_{\theta}^{Clayton}(u, v) = \left[\max(u^{-\theta} + v^{-\theta} - 1, 0) \right]^{-1/\theta} \quad (21)$$

This ensures that the expected loss reflects portfolios with concentrated physical risk, without distorting calculations during non-stressed baseline years.

3.5 Market Stranding and Uninsurability Thresholds

Physical damage to collateral is a component of the Loss Given Default (*LGD*) equation. Other considerations include market stranding (liquidity of the collateral) and uninsurability.

The engine introduces a decay function for uninsurability. Once the asset's Climate Risk Score surpasses an empirical threshold, the engine models a reduction in available insurance coverage. When the uninsurability threshold is breached, the insurance mitigation drops to zero. However, to translate the joint probability of the Copula into a physical loss severity, the engine computes a WWR Multiplier. This metric isolates the tail dependence by dividing the joint probability (the Copula output) by the probability of the events occurring independently:

$$WWR_Multiplier = \frac{C_{\theta}^{Clayton}(PD_{stressed}, P_{Damage})}{PD_{stressed} \times P_{Damage}} \quad (22)$$

This dependency ratio is applied as a scalar to the baseline physical collateral damage. To mathematically prevent the loss severity from exceeding the total value of the asset, the calculation is bounded at 1.0 (100% loss) prior to the application of the dynamic insurance threshold:

$$LGD_{stressed} = \min(1.0, LGD_{base} \times WWR_Multiplier) \times (1 - Insurance_Cover_t) \quad (23)$$

Ultimately, the primary goal of the preceding derivations is to compute the final Expected Credit Loss (*ECL*) for each loan under severe climate stress. While $PD_{stressed}$ and $LGD_{stressed}$ are heavily shocked by the complex climate manifolds, the bank's Exposure at Default (*EAD*) remains a relatively static accounting input (comprising drawn balances plus a standard credit conversion factor applied to undrawn credit lines), yielding the final loan-level loss metric:

$$ECL_{Stressed} = PD_{stressed} \times LGD_{stressed} \times EAD \quad (24)$$

4 Institutional Strategic Risk & Capital Degradation

While individual loan defaults drive portfolio losses, systemic climate events also influence strategic liquidity and market risks across the institution's balance sheet.

4.1 Unit-Aligned CET1 Capital Degradation

The engine scales micro-level exposures to match the macro-financial parameters of the bank (e.g., aligning EADs with the institution’s Rupee capital base). The system aggregates facility losses (ΔECL) and institutional stresses (Treasury Mark-to-Market losses, Liquidity Drawdowns) into $Loss_{Total}$.

The metric of institutional survivability is the Common Equity Tier 1 Capital Ratio variation:

$$CET1_{Stressed} = \frac{CET1_{CapitalBase} - Loss_{Total}}{RWA_{base}(1 + r_{inf})} \quad (25)$$

This metric informs the assessment of the bank’s capacity to absorb systemic climate shock relative to regulatory capital minimums.

4.2 Institutional Stress Variable Definitions

The components driving the stressed CET1 ratio are mathematically defined as follows:

4.2.1 Baseline Regulatory Inputs

The $CET1_{CapitalBase}$ and RWA_{base} are static accounting inputs derived from the institution’s core balance sheet prior to the injection of the climate shock:

- $CET1_{CapitalBase}$: The core equity tier 1 capital held in reserve to absorb losses.
- RWA_{base} : The aggregate value of the bank’s assets weighted by their historical baseline risk profiles.

4.2.2 Risk-Weighted Asset Inflation (r_{inf})

As the underlying credit quality of the loan portfolio deteriorates due to the realization of physical and transition hazards, regulatory capital requirements force a highly non-linear inflation of the portfolio’s Risk Weights. Rather than relying on a static or crude macroeconomic pass-through multiplier, the engine actively models this dynamic ratings migration as an emergent property of the portfolio.

For every loan in the simulated universe, the engine recalculates the precise capital requirement (K) under the Basel III Advanced Internal Ratings-Based (A-IRB) framework using the dynamically stressed Default Probability ($PD_{stressed}$) and Loss Given Default ($LGD_{stressed}$). The true systemic RWA inflation rate (r_{inf}) is explicitly derived by taking the ratio of the newly aggregated stressed RWA against the baseline RWA boundary:

$$r_{inf} = \frac{\sum_{i=1}^N [K(PD_{stressed,i}, LGD_{stressed,i}) \times 12.5 \times EAD_i]}{\sum_{i=1}^N RWA_{base,i}} - 1 \quad (26)$$

Where $K(\cdot)$ represents the standardized Basel III corporate asset correlation and maturity adjustment function. The 12.5 scaling factor is a global regulatory constant derived from the Basel minimum capital ratio of 8% (where mathematically, $RWA = \frac{MinimumCapital}{0.08}$, yielding the 12.5 multiplier to convert the raw capital requirement percentage K back into an equivalent Risk-Weighted Asset volume).

4.2.3 Total Institutional Loss ($Loss_{Total}$)

The total loss is an aggregate of five distinct stress vectors sweeping across the institution's balance sheet, transcending simple loan defaults:

$$Loss_{Total} = \Delta ECL_{Inc} + Liquidity_{Draw} + Market_{Loss} + Funding_{Cost} + Op_{Loss} \quad (27)$$

Where:

- ΔECL_{Inc} : The sum of all incremental Expected Credit Losses generated by borrower defaults ($\sum ECL_{Stressed} - ECL_{Base}$).
- $Liquidity_{Draw}$: Cash flow losses caused by distressed borrowers maximizing their undrawn credit lines prior to default.
- $Market_{Loss}$: Mark-to-market shock on the bank's treasury portfolio due to widening climate credit spreads.
- $Funding_{Cost}$: The increased premium the bank must pay in wholesale funding markets due to its own credit rating downgrade.
- Op_{Loss} : Direct physical damage to the bank's operational infrastructure and branch network, augmented by transition-related regulatory fines.